

ME315A_FarahLab03

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Lab 03

```
library(dplyr)
```

Anexando pacote: 'dplyr'

Os seguintes objetos são mascarados por 'package:stats':

filter, lag

Os seguintes objetos são mascarados por 'package:base':

intersect, setdiff, setequal, union

```
library(readr)
library(leaflet)
library(tidyr)
```

```
airlines <- read.csv("airlines.csv")
airports <- read.csv("airports.csv")
```

```
airlines <- read_csv("airlines.csv")
```

Rows: 14 Columns: 2

-- Column specification -----

Delimiter: ","

chr (2): IATA_CODE, AIRLINE

- i Use ``spec()`` to retrieve the full column specification for this data.
- i Specify the column types or set ``show_col_types = FALSE`` to quiet this message.

```
airports <- read_csv("airports.csv")|>
  select(IATA_CODE, CITY, STATE, LATITUDE, LONGITUDE)
```

Rows: 322 Columns: 7

-- Column specification -----

Delimiter: ","

chr (5): IATA_CODE, AIRPORT, CITY, STATE, COUNTRY

dbl (2): LATITUDE, LONGITUDE

- i Use ``spec()`` to retrieve the full column specification for this data.
- i Specify the column types or set ``show_col_types = FALSE`` to quiet this message.

```
processar_chunk <- function(chunk, pos) {
  chunk|>
    filter(!startsWith(as.character(DESTINATION_AIRPORT), "1"))|>
    drop_na()|>
    group_by(DESTINATION_AIRPORT)|>
    summarise(
      soma_atraso = sum(ARRIVAL_DELAY),
      n = n(),
      .groups = "drop"
    )
}

estatisticas_chunks <- read_csv_chunked(
  "flights.csv",
  callback = DataframeCallback$new(processar_chunk),
  chunk_size = 1000000,
  col_types = cols_only(
    DESTINATION_AIRPORT = col_character(),
    ARRIVAL_DELAY = col_double()
  ))
flights <- estatisticas_chunks|>
  group_by(DESTINATION_AIRPORT)|>
  summarise(
    media_atraso = sum(soma_atraso) / sum(n),
    .groups = "drop"
  )
```

2. a) A chave na tabela flights é "DESTINATION_AIRPORT"
b) A chave na tabela airports é "IATA_CODE"

```
# Identifica quais aeroportos de flights não estão em airports
aeroportos_ausentes <- flights|>
  anti_join(airports, by = c("DESTINATION_AIRPORT" = "IATA_CODE"))
print(aeroportos_ausentes)
```

```
# A tibble: 0 x 2
# i 2 variables: DESTINATION_AIRPORT <chr>, media_atraso <dbl>
```

```
flights <- flights|>
  left_join(
    airports|>
      select(IATA_CODE, CITY, STATE),
    by = c("DESTINATION_AIRPORT" = "IATA_CODE"))
```

- 3.

```
# Contagem de aeroportos por estado em ordem decrescente
aeroportos_p_estado <- airports|>
  group_by(STATE)|>
  summarise(qtd_aeroportos = n())|>
  arrange(desc(qtd_aeroportos))
print(aeroportos_p_estado)
```

```
# A tibble: 54 x 2
  STATE qtd_aeroportos
  <chr>         <int>
1 TX             24
2 CA             22
3 AK             19
4 FL             17
5 MI             15
6 NY             14
7 CO             10
8 MN              8
9 MT              8
10 NC             8
# i 44 more rows
```

- 4.

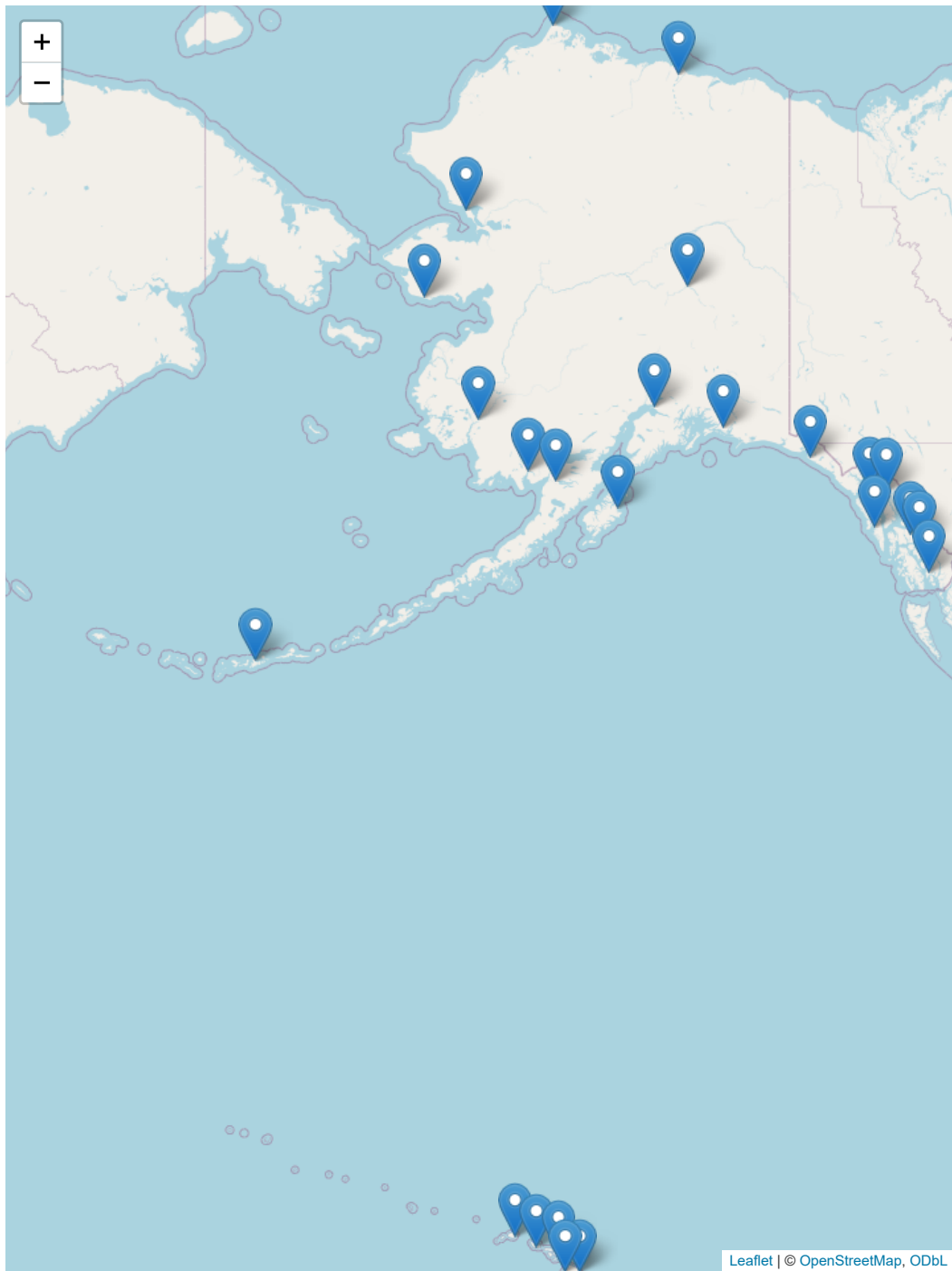
```

flights_mapa <- flights|>
  inner_join(
    airports|>
      select(IATA_CODE, LATITUDE, LONGITUDE),
    by = c("DESTINATION_AIRPORT" = "IATA_CODE")
  )|>
  drop_na(LATITUDE, LONGITUDE)

MAPA <- leaflet(flights_mapa)|>
  addTiles()|>
  addMarkers(
    lng = ~LONGITUDE,
    lat = ~LATITUDE,
    popup = ~paste0("<b>Aeroporto: </b>", DESTINATION_AIRPORT, "<br>",
      "<b>Cidade: </b>", CITY, " - ", STATE, "<br>",
      "<b>Atraso Média: </b>", round(media_atraso, 2), " min"))
MAPA

```

Warning in file.info(file): cannot resolve owner of file
 '\\smb\ra254775\Documentos\file4948597b47bd\webshot49487e872860.pdf': Não foi
 feito mapeamento entre os nomes de conta e as identificações de segurança



```
Sys.setenv(TZ = "America/Sao_Paulo")  
Sys.time()
```

```
[1] "2026-08-31 18:29:49 -03"
```